The Influence of Ozone: Superstoichiometric Oxygen in Atomic Layer Deposition of Fe_2O_3 Using *tert*-Butylferrocene and O_3

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Understanding the chemical mechanisms at play in atomic layer deposition (ALD) is critical for effective process development and expansion of ALD into more complex classes of materials. In this work, a mechanistic study of iron oxide deposited by ALD using *tert*-butylferrocene and ozone as reactants is performed. Iron oxide ALD using ozone is a useful model system for mechanistic studies due to the prevalence of ozone-based ALD processes and the uses of iron oxide in ternary and quaternary metal oxides. Results show that saturation conditions require significantly greater exposures of both reactants than is typically reported in the literature, and growths per cycle of greater than one monolayer of Fe_2O_3 per cycle are observed and explained. A growth mechanism is proposed whereby increased ozone exposure results in uptake of superstoichiometric oxygen into the film. X-ray characterizations reveal the presence of excess oxygen stored near the surface of films deposited with larger ozone exposures and show that increased ozone exposures cause crystalline domain rearrangement and conversion of the film from the γ-maghemite phase to the α-hematite phase. The mechanism described here has implications for the wider class of ozone-based ALD processes, and potential applications of this growth phenomenon are discussed.

1. Introduction

As nanotechnology pushes ever further into the current sphere of manufacturing technologies, it is critical to have an effective toolbox of processing techniques capable of fabricating structures at the nanoscale. Thin films in particular have proved useful in a variety of industrial settings, with applications for materials ranging from metals to oxides to nitrides and more.[1–4] These films can be deposited by a variety of methods, including physical processes like sputtering and evaporation and by chemical processes, such as sol–gel, chemical vapor deposition (CVD), and atomic layer deposition (ALD). ALD in particular is a powerful synthetic tool because the films deposited by this method possess excellent uniformity, can remain conformal in high aspect ratio structures, and yield highly controllable thickness modulation down to the ångström scale.[3–7] These properties are particularly advantageous in applications such as integrated circuits, memory, and photochemical devices that involve spatially compact or highly textured substrates.[2,8] To meet increasingly complex demands of material composition and properties, a wide range of materials has been synthesized by ALD including multicomponent and doped materials.[5,9] In order to control and tune the properties of these ALD materials, particularly in multicomponent ternary or quaternary materials, it is then important to have an accurate understanding of how the individual ALD processes themselves operate.

ALD metal oxides have proven useful in a variety of applications, including microelectronics, energy storage, catalysis, and photovoltaics.[1,5,8,9] Iron oxide in particular has demonstrated utility in thermal catalysis and electrocatalysis,[10–16] gas sensors,[17–19] and absorbers and electrodes for photoactive devices.[20–22] Ternary and quaternary materials containing iron have garnered strong interest for use as magnetic materials for microelectronics and information storage,[23–30] and have also been increasingly used as catalysts,[31–35] battery materials,[36–40] and photoactive devices.[34,41–43] Fe_2O_3 also shows promising performance as a photoanode, with carrier transport and recombination being sensitive to crystalline structure,[44,45] motivating an understanding of the relationship between synthesis and materials properties. For the iron oxide ALD process to be useful and implemented, particularly in multicomponent ALD, it is then critical to understand the process's growth behaviors and mechanisms. However, despite the wide applicability of iron oxide ALD materials, the mechanism of its growth is not well understood. In this work, ALD growth mechanisms of Fe_2O_3 using *tert*-butylferrocene (TBF) and ozone are studied. This liquid, functionalized metallocene precursor was chosen instead of the solid ferrocene because of its similar chemistry to ferrocene combined with its improved vapor delivery. While oxygen has been shown to be a sufficiently reactive coreactant for ferrocene, it has only been demonstrated at elevated ALD temperatures (above 350 °C), and the reports could not clearly confirm self-limiting growth;[46,47] therefore, ozone was used in this work because of its reactivity at lower temperatures. Iron oxide ALD from metallocene precursors and ozone has been

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reported previously, but existing reports were inconsistent in their growths per cycle and precursor exposures, and sometimes carbon impurities were evident in the films.[20,48,49] Furthermore, none of the previous work explored saturation conditions thoroughly, with one report performing a partial study of it but holding exposure fixed in a subsaturating regime,[48] and the others not addressing O_3 saturation altogether.[20,49]

The main aim of this work is to investigate the role of ozone in Fe_2O_3 ALD and its impacts on the growth process and resulting film properties. In this study, process characterization data are presented to explain inconsistencies in the literature and to gain a more complete understanding of the system. Special attention is given to the effect of ozone exposure on the growth per cycle (GPC), composition, and crystal structure of the deposited Fe_2O_3 films, and we show that superstoichiometric oxygen plays an important role in each of these properties, which has not been previously described. Ozone can be used to tune the resulting material properties without the need for high-temperature annealing, resulting trends are explained with a proposed growth mechanism, and applications of the growth mechanism and potential generalizations to wide classes of ALD systems are presented.

2. Results and Discussion

2.1. ALD Process

Due to the self-limiting nature of each ALD half-cycle, an ideal ALD process will exhibit saturated surface reactions without CVD side-reactions such that a plateau in GPC is observed with increasing precursor exposure.

Figure S1 (Supporting Information) shows the GPCs observed by variable-angle spectroscopic ellipsometry (VASE) as a function of TBF precursor exposure for various levels of ozone exposure. It can be seen that increasing ozone exposure leads to higher saturating GPCs, and that with higher ozone exposures more TBF is required to reach saturation. **Figure 1**a,b summarizes the saturated GPCs observed with increasing ozone and TBF exposures, respectively. Complete saturation studies at ozone exposures greater than 80 s per cycle were not performed due to the extensive processing time required by such large ozone and TBF exposures. The data show that reaching saturated growth requires at least 60 s of ozone exposure and 250 Torr-s of TBF exposure, resulting in a growth per cycle of ≈4 Å per cycle.

Hallmarks of ALD include its uniformity and constant growth rate with respect to number of cycles. Thickness across the stage holder was found to remain within 4% of the mean thickness, and across a single substrate to within 0.5%. As seen in Figure 1c, the growth per cycle is also maintained across a

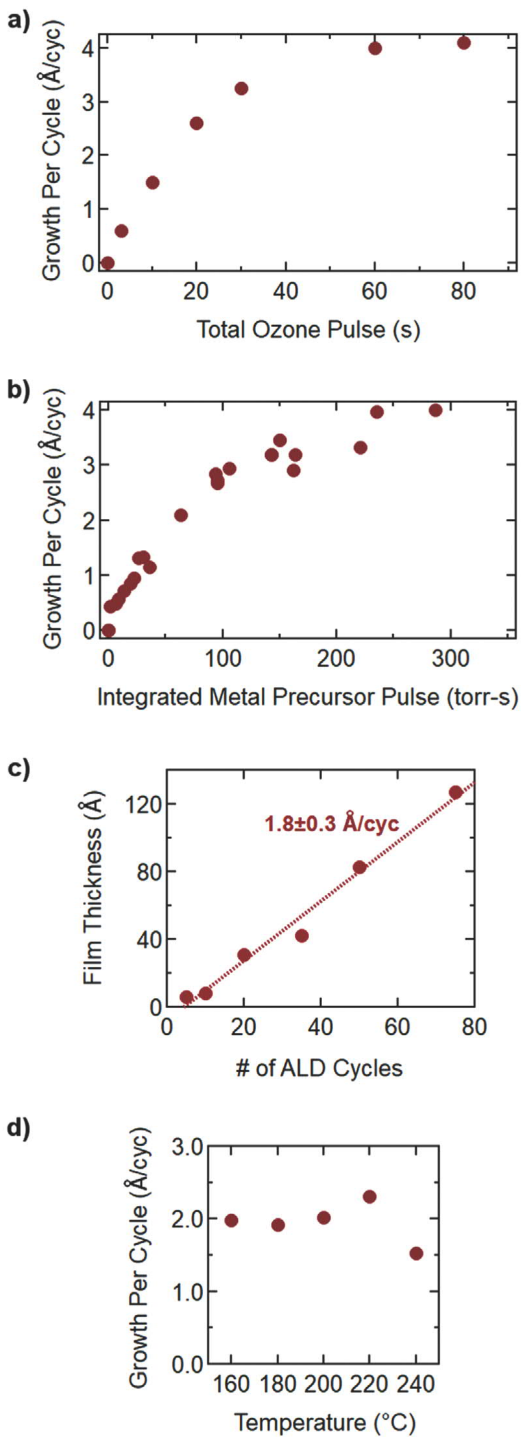

Figure 1. Growth of iron oxide films as measured by VASE. a) The growth per cycle as a function of ozone exposure with a saturating exposure of TBF. Because the TBF exposure required to reach saturation increases with ozone exposure, at each data point in (a), the TBF exposure was increased sufficiently to ensure TBF saturation for each ozone pulse length. b) The growth per cycle as a function of TBF exposure with a saturating exposure of ozone. Similar to (a), the ozone exposure used to generate each data point was increased sufficiently to ensure ozone saturation for each TBF exposure. c) The film thickness as a function of number of cycles. All films were deposited with 10 s of ozone and 90–100 torr-s of TBF per cycle (this TBF dose corresponds to saturation for this particular ozone exposure). The dotted line is the linear least squares best fit to the data, the indicated growth per cycle corresponds to the slope of this line, and the given error corresponds to a 95% confidence interval from a *t*-test. d) The growth per cycle as a function of substrate temperature. These depositions were performed with 30 s of ozone exposure per cycle for 25 cycles. Each data point in (a)–(c) represents one sample. Each data point shown in (d) is the mean of two separate measurements, except for the data at 180 and 200 °C which are each a single measurement.

range of cycle numbers. The slope of the line from a linear least squares fit indicates a GPC of 1.8 ± 0.3 Å per cycle for 10 s of ozone exposure per cycle, in agreement with Figure 1a. It is notable that this linear fit line does not intersect the horizontal axis at 0, but instead at 4.3 cycles, suggesting a short growth delay. Since ex situ VASE can be less reliable at low ALD cycle numbers when deposited material may be present in islands rather than a continuous film, additional characterization is needed to further probe this possible growth delay at low cycle numbers. Finally, Figure 1d illustrates that this linear growth per cycle also remains relatively constant over temperatures between 160 and 220 °C. A lower growth per cycle is observed at higher temperatures, which may be attributed to increased ozone decomposition.

The saturated GPC of 4 Å per cycle is higher than is typically observed in ALD processes,[1,5,9] and the exposures of both TBF and ozone required to reach saturation are larger than most previous reports on the process.[20,48,49] The high GPC and large precursor exposures required to reach saturation suggest that the growth mechanism of this process may be more complex than a simple surface-limited reaction that deposits a monolayer or less of material per cycle.

2.2. Film Density and Composition

To probe the phenomenon behind this growth behavior, composition and density of the films were investigated. X-ray photoelectron spectroscopy (XPS) measurements revealed carbon concentrations within the film typically ranged from 3% to 6%, comparable to previous reports,[48,50] but the low signal-to-noise ratio prevents a more detailed analysis of the C1s spectra (see Figure S2, Supporting Information). Figure S2 (Supporting Information) also indicates that even after GCIB sputtering, most of the carbon remaining in the film is present near the surface. Therefore, the actual bulk carbon concentration is likely less than the overall 3–6% concentration we measure. XP spectra, such as the one shown in Figure S3 (Supporting Information), of the as-deposited films give an Fe $2p_{3/2}$ peak that also corresponds well to previous reports of the Fe_2O_3 phase, including the complex multiplet splitting and satellite structure.[51] Because the measured growth per cycle is greater than the roughly 2.3 Å reported thickness of a monolayer of Fe_2O_3,[52,53] this indicates that the process deposits more than one full monolayer of Fe_2O_3 per ALD cycle. Angle-resolved XPS (AR-XPS) measurements were also performed on samples with different ozone exposures. Analysis of the relative oxygen concentration from the AR-XPS measurements (**Figure 2**) reveals an interesting observation: whereas the composition of the films in the bulk corresponds to Fe_2O_3, the surface region of the film contains excess oxygen beyond the 60% relative composition expected from Fe_2O_3. Furthermore, the amount of excess oxygen present correlates with the ozone exposure, suggesting that increasing ozone exposure introduces superstoichiometric oxygen in the region of the film near the surface. The superstoichiometric oxygen does not seem to be stable for extended periods of time in ambient conditions, however, since after 13 months in air at room temperature the gradient in oxygen composition near the surface is no longer detected (see Figure S4, Supporting Information), and the O 1s XPS peak structure is constant with respect to angle (see Figure S5, Supporting Information).

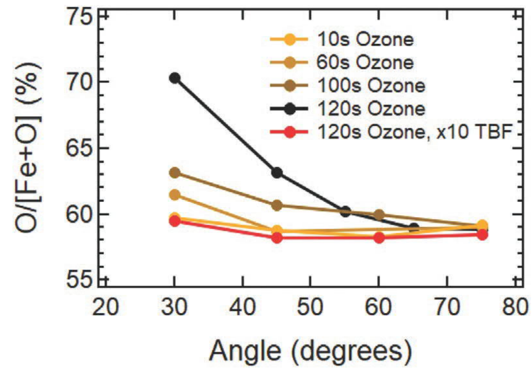

Figure 2. Film compositions calculated from angle-resolved XPS measurements of Fe_2O_3 films grown with varying levels of ozone exposure. All samples were grown with 20–50 torr-s of TBF exposure except for the x10 TBF film which was grown using 400 torr-s TBF exposure. Film thicknesses are 160, 135, 215, 50, and 90 Å, in order of legend appearance. Angles are given with respect to the sample surface plane, so smaller angles are at more grazing incidence and thus more surface sensitive.

Analysis of the Fe 2p and O 1s XPS peaks as a function of angle for both high and low ozone exposures, as shown in **Figure 3**, can give further insight into the chemical states of these species. It is apparent in Figure 3a,b that although the intensities of the peaks increase with angle as more photoelectrons are collected from deeper within the film, the overall peak shape and structure of the Fe 2p feature remain constant. This result implies that the oxidation state and chemical environment of the iron atoms are similar in the surface region and in the bulk for both 10 and 120 s of ozone exposure. However, Figure 3c,d indicates that while the chemical state of oxygen is homogeneous throughout the film at a 10 s exposure, this is not the case at a 120 s exposure when excess oxygen is present in the film. The peak near 532 eV, which is usually associated with defect sites or with hydroxyl, oxyhydroxyl, or peroxyl species,[54–58] appears more strongly under high ozone exposure. Furthermore, this peak has greater relative intensity at shallower incidence angles under conditions of 120 s ozone exposure. The spectral behavior indicates that these oxygen species are more prevalent near the surface of the film and suggests that the peak near 532 eV is associated with the superstoichiometric oxygen present near the surface of the film.

The different properties of the surface region of the film can also be observed using X-ray reflectivity (XRR). In XRR, the critical angle at which the initial drop in intensity occurs is determined by the electron density at the surface of the film, and the electron density in the bulk determines the overall decay rate of the reflectivity with increasing angle.[59,60] Multiple film models were tested in order to find the best fit to the measured XRR intensity. It was found that modeling the film with a constant, uniform density was not sufficient to fit both the critical angle and the overall decrease of intensity with angle (see **Figure 4**a). In contrast, modeling the film with a gradient of density is effective at capturing all features of the experimental data. This model also maintains its efficacy with films of other thicknesses, one of which is shown in Figure S6

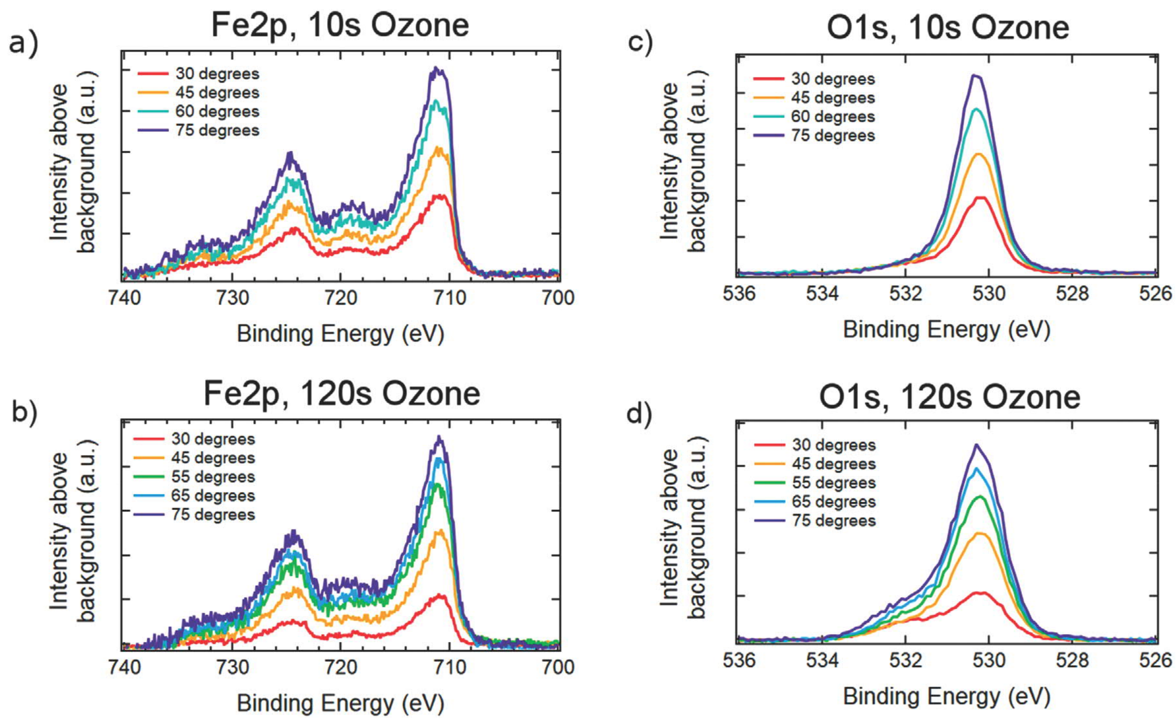

Figure 3. Angle-resolved XPS measurements of Fe_2O_3 films grown with ≈20 torr-s of TBF exposure per cycle. a,b) depict the Fe 2p peak, and c,d) show the O 1s peak. The sample shown in (a) and (c) was grown using 10 s of ozone exposure to a thickness of 150 Å, and the sample measured in (b) and (d) was grown with 120 s of ozone exposure to a thickness of 50 Å.

(Supporting Information). The XRR model was constructed using a bilayer of two materials of different densities with high interfacial mixing, and it indicates the near-surface region of the film has a lower density than that of the bulk. As seen in Figure 4b, the interfacial mixing between the region of excess oxygen and the bulk of the film does not result in a sharp, well-defined boundary but instead a gradual change in density. Typical values for surface density in the model were found to be 3.5–4.0 g cm^{-3}, and typical values for the bulk density were 5.0–6.0 g cm^{-3}, near the expected density of 5.3 g cm^{-3} for α-Fe_2O_3.[53] Surface roughness extracted from the XRR fits also corresponds well with direct atomic force microscopy (AFM) measurements (shown in Figure S7, Supporting Information), further corroborating use of the selected model.

2.3. Film Growth Mechanism

The observation that the surface of the film takes up superstoichiometric oxygen with increasing ozone exposure suggests a cyclical mechanism of ALD growth. A proposed model is illustrated in **Figure 5**, whereby oxygen is iteratively taken up into the film from ozone and then withdrawn and consumed by TBF. Such a mechanism would require the ability of oxygen to be transported through the film, and indeed oxygen vacancies and mobility through Fe_2O_3 are well-documented phenomena.[61–64] A previous study by Martinson et al. used in situ quadropole mass spectrometry to examine the reaction byproducts between ferrocene (a precursor chemically similar to TBF) and ozone.[48] In the work, they found that ferrocene adsorbed to the surface, reacting with surface oxygen species to release cyclopentadiene and cyclopentadienone byproducts. They also observed CO_2 and H_2O as byproducts of the reaction of O_3 with the surface, characteristics of hydrocarbon ligand combustion. These findings indicate consumption of surface oxygen species in the adsorption of the organometallic precursor, and replenishment of surface oxygen species and removal of carbon with ozone. If oxygen is capable of being stored subsurface and is sufficiently mobile to reach the surface, both of these hypothesized molecular reactions could continue to progress even after a single monolayer of precursor is deposited, leading to more than one monolayer of material deposited each cycle. Unfortunately, their study was unable to obtain a quantitative analysis of surface chemistry or kinetics, so additional study here is needed to elucidate the molecular mechanisms at play.

The proposed mechanism is consistent with the growth behaviors observed in Figure 1: if the film can act as a reservoir for oxygen then the ALD half reactions would no longer be confined to the surface monolayer of the growing film, resulting in higher saturated GPCs. This growth model also explains the inhomogeneities in both composition and density found in the film and is consistent with the saturation behavior observed for the process. Additionally, the increased growth of more than a monolayer each cycle would require additional reactant exposure supplied to the surface to reach saturation, particularly if it is needed to allow for the physical diffusive mechanism of oxygen uptake and consumption into and out of the film. Indeed, both larger saturation exposures and higher growth per cycle are indicated by the data in Figure 1.

The proposed growth model is further supported by an additional experiment performed in which a long ozone exposure is paired with an extra-long TBF exposure. The resulting data are

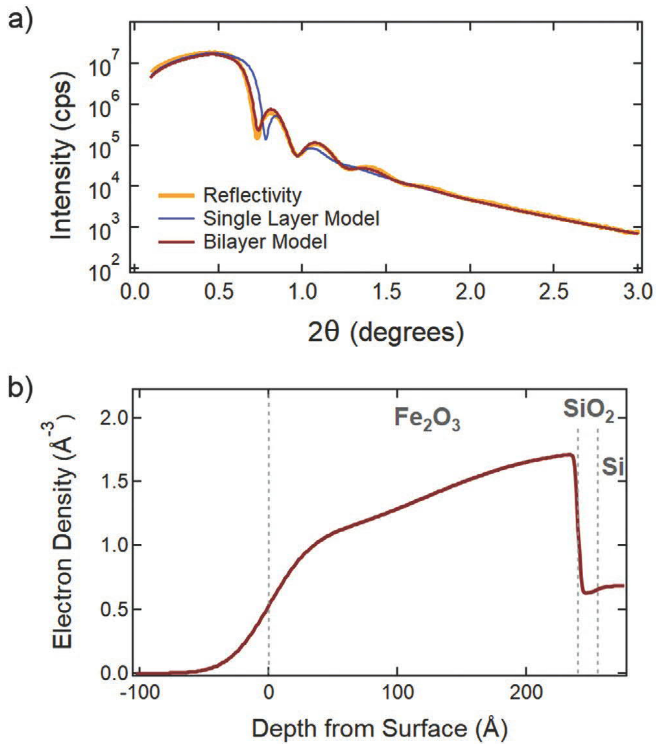

Figure 4. a) XRR measurement of a 24.0 nm thick Fe_2O_3 film, shown in orange. Calculated intensity shown in red is generated from a bilayer model reflecting a gradient of density within the film, in which the surface region of the film is less dense than the bulk. The blue curve is generated from a simpler model where density is assumed to be constant throughout the film. Data are truncated below $\theta = 0.04°$ to omit sample edge effects from the analysis. b) A plot of the electron density of the bilayer model used in (a), reflecting the thicknesses, densities, and roughnesses found from fitting the model in (a). The lower density region of excess oxygen is visible near the surface with a gradual transition to the bulk Fe_2O_3 region, followed by the SiO_2/Si substrate of lower electron density. The smooth contours at the surface and at interfaces arise from surface and interfacial roughness.

shown in red in Figure 2. The results indicate that increasing the exposure of TBF causes most of the excess oxygen present in the film to be consumed. The data may reveal the presence of a small amount of excess oxygen even after the increased TBF exposure; this could have resulted from insufficient precursor exposure as we expect that if insufficient TBF is provided to consume all the excess oxygen, some oxygen will remain in the film. The observation that the excess oxygen can be removed suggests the superstoichiometric oxygen present in the film is in a reactive state, capable of being consumed by reaction with TBF, and that it is mobile enough such that it can be expended with sufficient TBF exposure.

2.4. Crystallinity Modulation

A physical mechanism of oxygen moving through the film may be expected to have structural effects on the film as it grows, so grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were performed to explore its crystallographic phase. Three measurements of representative Fe_2O_3 films synthesized using varying ozone exposures are shown in **Figure 6**a. Each of the resulting films exhibits peaks characteristic of the α-hematite phase, the γ-maghemite phase of Fe_2O_3, or both. These phases will henceforth be referred to as the α and γ phases. It can be seen that increasing ozone exposure enhances overall crystallinity, indicated by the increased diffraction peak intensities. Scanning electron microscopy (SEM) images taken of Fe_2O_3 grown with a range of ozone exposures corroborate the presence of enhanced polycrystalline domain formation with higher ozone exposure. Three such images are shown in Figure S8 (Supporting Information). Besides an overall increase in diffraction peak intensity with higher ozone exposure, there is a change in the relative intensity of peaks associated with the α-Fe_2O_3 versus γ-Fe_2O_3 phase with increasing ozone. In particular, at lower ozone exposures the diffraction pattern is consistent with the presence of the γ-Fe_2O_3 phase, but with increasing ozone exposure the diffraction peaks associated with the α-Fe_2O_3 phase grow in intensity. This result indicates that the film is gradually converted into α-Fe_2O_3 with increasing ozone exposure, an effect which can be quantified by measuring the respective peak areas. Relative phase compositions were calculated using the approach described in the Experimental Section, and Figure 6b illustrates the results of these calculations. The data show that the phase of the Fe_2O_3 thin film can be tuned by adjusting the ozone exposure during the ALD cycles, with the γ-Fe_2O_3 phase dominating at lower exposures and the α-Fe_2O_3 phase becoming more prominent at higher exposures.

In addition to the crystalline phase, crystalline domain orientation can also be tuned with ozone exposure. 2D GIWAXS patterns of α-Fe_2O_3 films grown using different ozone exposures are shown in **Figure 7**. The faint pentagonal pattern common to each subfigure arises from the silicon substrate and indicates that the measurements shown in Figures 6 and 7 probe the entirety of the Fe_2O_3 film. In Figure 7a, at an intermediate exposure of 30 s of ozone, rings associated with the α-Fe_2O_3 diffraction peaks can be seen on top of the pentagonal background scattering pattern from the silicon substrate. A faint ring can be seen just above 2 Å^{-1} corresponding to the first γ-Fe_2O_3 diffraction peak, and all other peaks in Figure 7a–d correspond to α-Fe_2O_3. The rings associated with the γ phase are not visible in Figure 7b–d due to their weak relative intensity and the films' conversion to the α phase as described in Figure 6b. The rings associated with the α phase in Figure 7a have mild intensity modulations with respect to angle, indicating somewhat random crystallite orientation with some degree of preferential orientation. In Figure 7b, it can be seen that an increased ozone exposure of 60 s increases the strength of the scattering pattern rings. Furthermore, as illustrated in more detail in Figure S9 (Supporting Information), the nonuniformity of the rings is also slightly more tightly concentrated. Spots of high intensity are visible in several of the rings, indicating the presence of texturing. The preferential crystalline domain orientation shown in Figure 7b shows i) the orientation of the (006) peak at $Q = 2.74$ Å^{-1} in the out-of-plane direction, perpendicular to the substrate, and ii) the (110) peak at $Q = 2.49$ Å^{-1} in the in-plane directions.[65] These observations indicate an orientation of the *c*-axis perpendicular to the substrate. Interestingly, further increasing the ozone exposure does not continue this

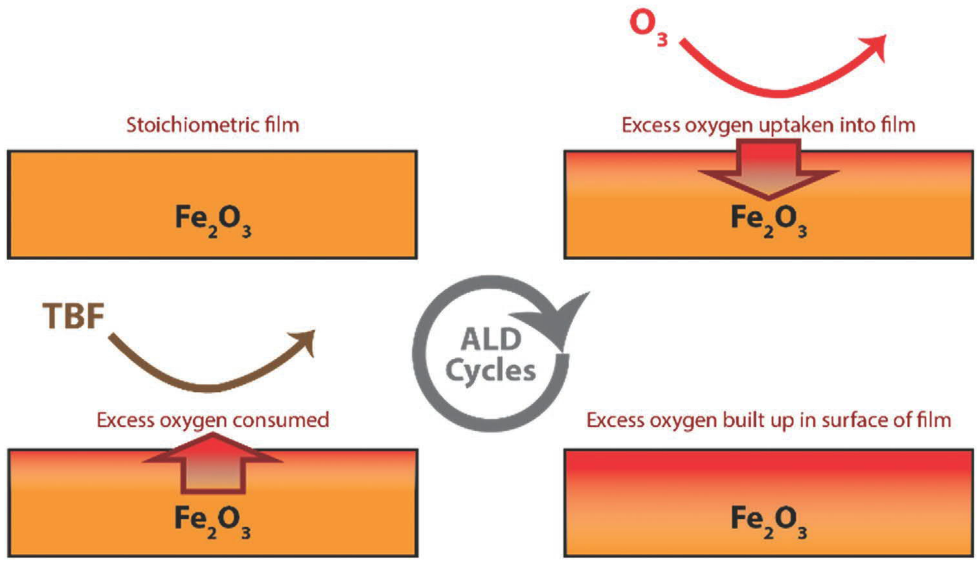

Figure 5. The proposed mechanism of Fe_2O_3 film growth during each ALD cycle, whereby each ozone exposure step introduces oxygen into the surface region of the film and each TBF exposure consumes oxygen from the film. This schematic is illustrated in the case in which the TBF exposure is saturating, so that all excess oxygen is consumed.

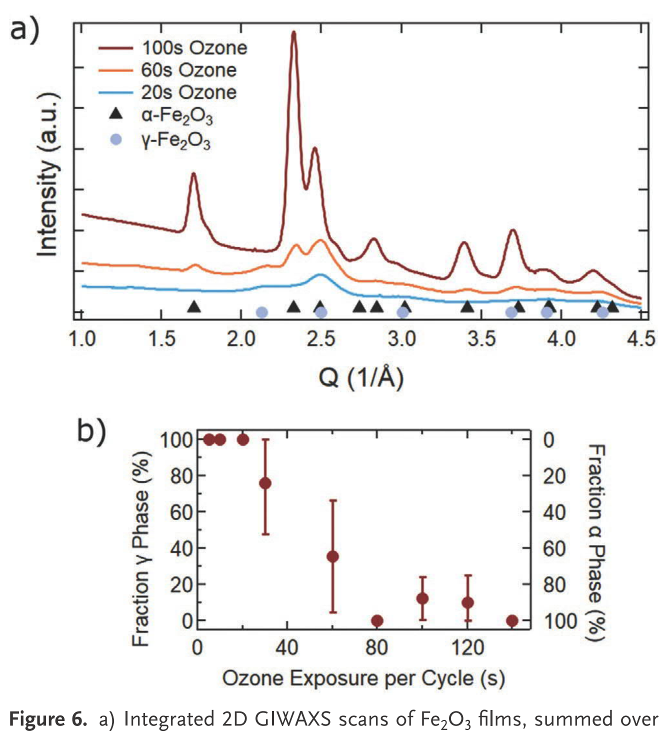

Figure 6. a) Integrated 2D GIWAXS scans of Fe_2O_3 films, summed over all measured angles χ as a function of Q. Three scans of films grown using 100, 60, and 20 s of ozone exposure under saturated TBF doses are shown, together with the primary peaks of α-Fe_2O_3 and γ-Fe_2O_3 indicated with black triangles and gray circles, respectively.[65,71] Film thicknesses for the 20, 60, and 100 s ozone samples are 90, 138, and 140 Å, respectively. b) Approximate phase composition between the α and γ phases, as calculated from the ratio of the area of the 2.13 $Å^{-1}$ peak to the area of the 2.49 $Å^{-1}$ peak. Films here were grown using a wide range of TBF exposures. Error bars indicating one standard deviation are shown for ozone exposures at which multiple samples were grown; points shown at 10, 20, 80, and 140 s consist of a single sample.

behavior. In Figure 7c, a high ozone exposure of 120 s is shown to completely disrupt the crystallinity of the film such that the only diffraction peaks visible by GIWAXS are produced by the underlying silicon substrate. This loss in crystallinity also corresponds to the presence of superstoichiometric oxygen in the film as seen by AR-XPS in Figure 2. Figure 2 indicates the surface region of the film contains an O:Fe ratio of 2.3:1, which is a 50% increase in oxygen content as compared to the 1.5:1 ratio that arises from the atomic arrangement in a standard α or γ phase. It is expected that the crystalline structure would be disrupted to accommodate such a significant increase in oxygen. It remains possible that the bottom of the film retains its crystalline structure but that the crystalline portion of the film is too thin to be detectable by the GIWAXS shown in Figure 7c. If, however, the TBF exposure is doubled to consume some of this excess oxygen, then some of the crystallinity is seen to recover, as shown by the presence of weak α phase diffraction rings in Figure 7d.

These GIWAXS measurements summarized in Figure 7b–d support a mechanism of cyclic oxygen uptake and consumption in the film. The crystallinity observable in Figure 7b for 60 s ozone exposure is disrupted by the time the ozone exposure is increased to 120 s, as shown in Figure 7c, by the physical process of oxygen moving into the film. When additional TBF is used (Figure 7d), the crystallinity is partially recovered, indicating that some of the disruptive oxygen has been removed and consumed. This is consistent with the phenomenon observed Figure 2 by XPS and explained by the model in Figure 5. We speculate that the process of repeatedly introducing and removing this disruptive oxygen allows the atoms in the film to rearrange, forming more energetically favorable structures in a process akin to annealing. This rearrangement with increasing ozone exposure would then result in the formation of the more thermodynamically stable α phase[66,67] instead

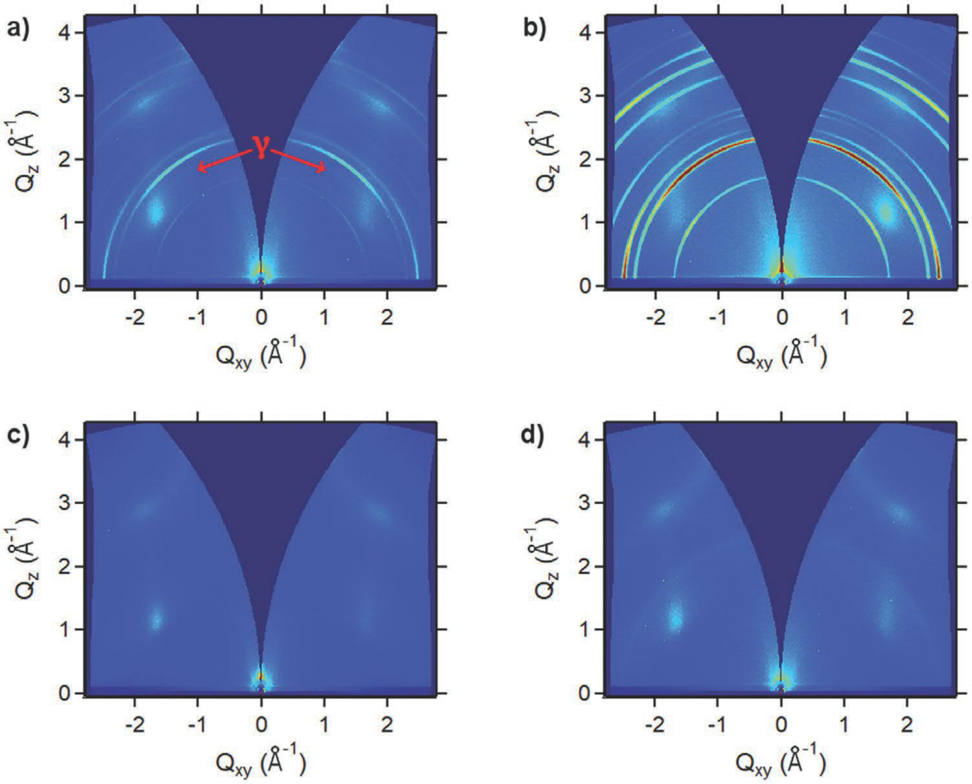

Figure 7. 2D GIWAXS patterns of Fe_2O_3 films grown with a) 30s, b) 60s, c) 120s, and d) 140s of ozone exposure per cycle. The TBF exposure in (d) is twice that of (c), and all patterns are plotted on the same intensity color scale. All visible diffraction rings correspond to the α phase with the exception of one faint γ phase ring in (a) that is indicated with the red arrows. The thicknesses of the films shown in (a–d) are 129, 135, 50, and 120 Å, respectively. The faint pentagonal pattern common to each subfigure is attributed to the silicon substrate.

of the γ phase, producing the trend observed in Figure 6b. Previous studies of α-Fe_2O_3 have found that oxygen vacancy defects are common[61,62] and that diffusion along the *c*-axis of the unit cell is much higher than that of diffusion perpendicular to the *c*-axis.[63,64] As observed in Figure 7, with increased oxygen exposures this axis is preferentially oriented perpendicularly to the substrate. It is interesting to note that the films reorganize such that the direction of most facile oxygen diffusion is oriented in the direction of oxygen movement as it is transported in and out of the film during each cycle.

2.5. Growth Mechanism Consequences

Overall, these GIWAXS experiments corroborate the compositional and density measurements obtained by AR-XPS and XRR indicating the presence and behavior of a superstoichiometric oxygen layer. As illustrated in Figure 2, its dependence on ozone exposures suggests ozone is involved in the formation of the superstoichiometric oxygen. Both AR-XPS and GIWAXS measurements also indicate that this oxygen can be consumed and removed with additional organometallic precursor. Together, these observations are explained by a growth mechanism involving iterative introduction and consumption of superstoichiometric oxygen, illustrated in Figure 5. This mechanism would result in structural changes to the film which are indeed observed in both SEM and GIWAXS. Finally, preliminary data (see Figure S10, Supporting Information) show NiO ALD using nickelocene and ozone exhibits similar growth behaviors to the Fe_2O_3 ALD process described in this work, suggesting this growth mechanism may generalize to other metal oxide ALD processes using ozone as a coreactant.

Moving forward, the growth mechanism has several implications. First, the physical movement of oxygen through the film with each ALD cycle allows the atoms to rearrange. In doing so, they produce a more stable structure, both in forming the α hematite phase and in orienting crystalline domains in such a way that maximizes the directional diffusivity of oxygen through the film. This behavior allows for the ozone coreactant to be useful as a tool for tuning the mechanical properties of the resulting films without high-temperature annealing. This could impact device usage of Fe_2O_3, such as in a photoanode, where recombination and carrier concentration are sensitive to grain boundaries and crystalline phase. Additionally, in applications like microelectronics in which capping layers are deposited on top of Fe_2O_3 films, it may be important to consider the layer of superstoichiometric oxygen; as the layer is seen to dissipate with time, this oxygen could form bubbles or other defects within devices that impact properties and performance.

Second, the oxygen uptake phenomenon may be related to that of a recent report of Fe_2O_3 ALD on Pt surfaces where it was hypothesized that oxygen was transported through the film.[68] In the study by Singh et al., it was observed that when performed on platinum substrates, iron oxide ALD could be achieved using *tert*-butylferrocene and molecular oxygen gas, obviating the need for ozone. While O_2 was not reactive enough to react with the iron precursor at moderate temperatures, it was hypothesized that the oxygen species were able to migrate through the film to the underlying platinum substrate where it was dissociated into reactive oxygen atoms to participate in the ALD reactions. The phenomena observed in the present work corroborate some of the findings of the platinum-activated study, in particular the ability of the iron oxide film to store oxygen species to result in increased growths per cycle.

A third potential use of this growth mechanism relates to multicomponent ALD. It is shown in this work that if the organometallic precursor exposure is chosen such that it is insufficient to consume all of the superstoichiometric oxygen present in the film, some of this oxygen persists in the film. When the Fe_2O_3 ALD process is then combined with a second metal oxide process in multicomponent ALD, those persisting reactive species could be available for reaction with the second metal precursor, thereby increasing the growth of the second metal oxide. This could have the effect of enhancing growth of the other metal oxide whenever it is grown on top of an iron oxide surface,[69] a practice common in the supercycle strategy of ternary ALD processes.[9] Such an enhancement could potentially be exploited for ALD of some ternary materials where one of the constituent oxides is challenging to grow; by introducing these reactive oxygen species into the film, otherwise unreactive precursors might be used to grow material on top of the iron oxide. Many ternary oxide processes that use ozone as a coreactant exhibit nonideal growth behaviors,[9] and it is possible that persisting oxygen species similar to those observed in this work are a contribution to these deviations.

3. Conclusion

An ALD process for the growth of Fe_2O_3 was developed and characterized, using *tert*-butylferrocene as the metal precursor and ozone as the coreactant. Saturation conditions and growth behavior were investigated, with an emphasis on understanding the effects of ozone exposure on the deposition process. A model for oxygen uptake into the film is presented, supported by X-ray spectroscopic and reflectivity measurements showing the presence of excess oxygen in the surface region of the film. The mechanism identifies and explains how high precursor exposure saturation conditions result in high ALD growths per cycle, an effect that has not yet been fully addressed in the literature. Implications of this growth mechanism are also reflected in the structural properties of the films. Ozone exposure was found to impact the crystallinity of the resulting films by both increasing film crystallinity and converting the γ maghemite phase into the α hematite phase. This phenomenon presents a novel way of rearranging atoms in the film without the need for higher temperature annealing. The phase dependence on ozone exposure can be used both to tune the phase between the pure γ phase and pure α phase and to control the orientation of crystalline domains. Other ALD processes relying on ozone as a coreactant may also exhibit similar behaviors, possibly including our preliminary work on NiO ALD and the study of MnO ALD by Young et al.[70] Additionally, this mechanism may affect growth and behavior of ternary ALD systems by introducing reactive surface oxygen species that could influence the precursor chemisorption. Further investigation of these phenomena and more detailed mechanistic studies are the subject of future work.

4. Experimental Section

Synthesis of Iron Oxide Films: All depositions of thin films were performed in a custom-built, perpendicular-flow ALD reactor with a 4 in. heated stage and substrate holder. Precursors were introduced through a showerhead above the stage. The reactor was purged with nitrogen gas at a flow rate of 50 SCCM, resulting in a reactor working pressure of 750 mtorr. Sample substrates were n-doped (100) silicon substrates with native oxide (WRS Materials) and were cleaned for 15 min in a Novascan PSD Series Digital UV Ozone System prior to deposition. The precursors used for the deposition were *tert*-butylferrocene [Fe(Cp)(Cp^{t}Bu), TBF, Strem Chemicals, 98%] and ozone generated from oxygen gas (Praxair, 99.6%) with an IN USA AC-2025 generator. Ozone concentration was measured by an IN USA Mini-HiCon Bench analyzer, and concentrations were kept in the range of 180–220 grams per normal cubic meter. The substrate temperature was held at 220 °C for all depositions unless otherwise noted, and the reactor walls were heated to 185 °C. The TBF bubbler was heated to 80 °C and the ozone line was at room temperature.

The reactor pressure was monitored using a convectron gauge, and the partial pressure of ozone during ozone pulses was constant at 1.1 Torr. Since the partial pressure of ozone was constant across experiments, the total ozone exposure is directly proportional to the total ozone pulse time, and the total ozone pulse time is adequate to quantify ozone exposure. To prevent the pump pulling vacuum on the ozone generator, ozone exposures of longer than 10 s were broken up into repeated pulses of 10 s separated by purge times of 30 s to allow the ozone delivery line to repressurize. In contrast, the partial pressure of TBF introduced to the reactor depended on pulse time and varied across experiments. To account for this variation, the total TBF exposure is quantified by approximating the integrated pressure-time area under the precursor pressure pulses during the pulse and soak times. To increase the total exposure of TBF to the substrates, the soak time of TBF was increased by introducing a pump block using the following protocol. Before the TBF pulse, both the pump and N_2 purge were isolated from the reactor chamber and remained isolated for the duration of the pulse. Following the TBF pulse, the reactor chamber was kept isolated from both the pump and N_2 purge gas for an additional soaking time before both the pump and purge gas were reintroduced simultaneously to purge the TBF from the chamber.

Physical Characterization: Variable-angle spectroscopic ellipsometry, used to determine film thickness, was performed on a Woollam Co. α-SE spectroscopic ellipsometer at 65° and 70° angles with a photon energy range of 1.39–3.25 eV. Film thicknesses were modeled in Woollam Co. CompleteEASE software using a general oscillator layer model containing a Tauc-Lorentz and a Gaussian absorption component on top of layer models for the native oxide and silicon substrate. The model was generated from thickness measurements performed using XRR across a range of thicknesses between 0 and 25 nm to ensure accuracy of the model for the film thicknesses used in this work. XRR was performed on a PANalytical X'Pert PRO X-ray diffractometer to characterize film thickness, roughness, and density. A Cu K-α radiation source was used to collect data in a 2θ range from 0° to 8°, and the data were modeled using PANalytical X'Pert Reflectivity software. Electron density was calculated by normalizing the scattering length density to that of silicon.

XPS on a PHI 5000 VersaProbe III with monochromated Al K-α X-ray source was used for elemental compositional analysis. Adventitious carbon surface species were cleaned in situ using an Ar 2500+ gas cluster ion beam (GCIB) at 2.5kV for 4 min prior to XPS. As shown in

Figure S11 (Supporting Information), this exposure was chosen due to observations that the carbon concentration showed a sharp decrease for the first ≈4 min of GCIB exposure before plateauing, suggesting that surface adventitious carbon species could be removed with 4 min of sputtering. Although Figure S2 (Supporting Information) suggests the 2.5 kV beam energy is not sufficient to remove all surface carbon species, a higher beam energy was not used so as to avoid altering the properties of the film. Peak fitting was performed using MultiPak software using a Shirley background. AFM was performed on a Park NX10 instrument, using a MikroMasch NSC35 tip over a 1 µm × 1 µm area. Finally, SEM was performed using an FEI XL30 Sirion scanning electron microscope with a field emission gun source, using a 5 keV incident electron energy and an in-lens detector to measure secondary electrons.

GIWAXS was performed at the Stanford Synchrotron Radiation Light Source (SSRL) at the SLAC National Accelerator Laboratory using a beam wavelength of 0.976 Å, a 2D Rayonix MX225 detector, a 0.2° incidence angle, and a 150 mm detector distance. 1D data were obtained from these 2D scans by summing across all values of χ for a constant value of Q. These summations were then used to calculate relative phase compositions using the following methodology. The two most prominent peaks in the α-Fe_2O_3 hematite phase at Q values of 2.33 and 2.49 $Å^{-1}$ correspond to the (104) and (110) crystal planes,[65] respectively, whereas according to ICCD card 00-039-1346, γ-Fe_2O_3 maghemite exhibits two peaks in that region at 2.13 and 2.49 $Å^{-1}$, corresponding to the (220) and (311) planes, respectively. Because the ratio between the sizes of these two maghemite peaks has been well-documented,[65] the size of the gamma phase-specific 2.13 $Å^{-1}$ peak can be used to determine the contribution of the gamma phase to the 2.49 $Å^{-1}$ peak area. Any remaining area in the 2.49 $Å^{-1}$ peak can then be assigned to the α-hematite phase since both phases share this peak. From this analysis, an approximation of the relative phase composition can be calculated by finding the percentage contribution of each phase to the area of the 2.49 $Å^{-1}$ peak.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

atomic layer deposition, iron oxide, ozone, surface modification, thin films

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